

Sheaves & Proofs

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Desired background

Some exposure to proof theory (sequent calculus)
and semantics (structures, satisfaction, soundness, completeness)
for classical predicate logic.

Not required background

- Intuitionistic logic (sequent calculus, Kripke models)
- Basic category theory (categories, functors, natural transformations)
- Basic point-set topology (topologies, covers)

Part 1

Coverage Semantics over Partial Orders

(The logic of localic toposes)

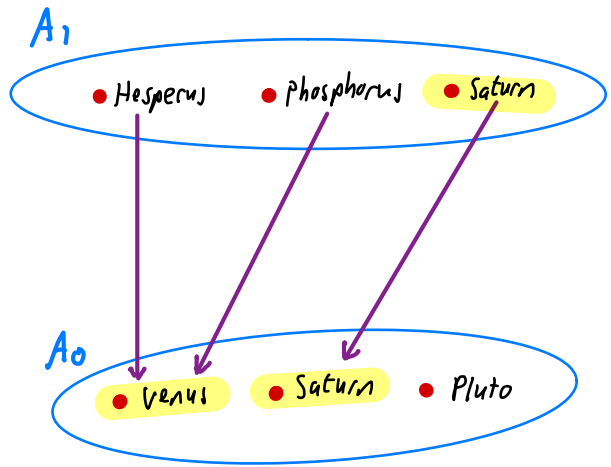
- Kripke semantics
- Coverage semantics
- Cut-elimination via constructive completeness
- The **77**-dense coverage.

(Inverted) Kripke semantics

A Kripke Model is given by

- A partial order $(W, \leq)$ of Worlds
- To every $w \in W$ a set A_w
- Transition functions $(t_{vw} : A_w \rightarrow A_v)_{v \leq w}$
Satisfying $t_{ww} = id_{A_w}$; $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- For every predicate P of arity k
Subsets $P_w \subseteq (A_w)^k$ satisfying
 $v \leq w \Rightarrow t_{vw}(P_w) \subseteq P_v$.

$$W = \begin{array}{c} 1 \\ | \\ 0 \end{array}$$



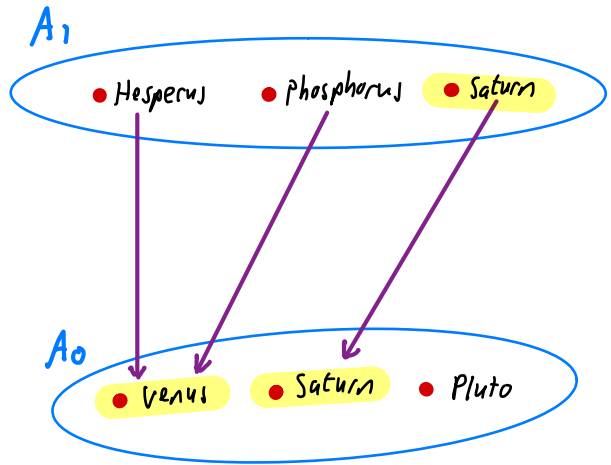
 The planet predicate

Category-theoretic Formulation

A Kripke model is given by

- A partial order $(W, \leq)$ of Worlds
- A functor $A: W^{op} \rightarrow \underline{\text{Set}}$
(i.e., a presheaf $A \in \text{Psh}(W)$)
- For every predicate P of arity k
a subpresheaf $P \subseteq A^k$

$$W = \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} 1 \\ \\ 0 \end{array}$$



 The planet predicate

Notation

$\text{Venus} = \text{top}(\text{Hesperus})$	(previous slide)
$= A(0<1)(\text{Hesperus})$	(this slide)
$= \text{Hesperus} \cdot 0$	(henceforth)

Kripke forcing relation

$$w \Vdash \phi$$

$$w \Vdash P(a_1, \dots, a_k) \Leftrightarrow (a_1, \dots, a_k) \in P_w$$

$$w \Vdash a_1 = a_2 \Leftrightarrow a_1 = a_2$$

$$w \Vdash \phi \wedge \psi \Leftrightarrow w \Vdash \phi \text{ and } w \Vdash \psi$$

$$w \Vdash \phi \vee \psi \Leftrightarrow w \Vdash \phi \text{ or } w \Vdash \psi$$

$$w \Vdash \perp \Leftrightarrow \text{false}$$

$$w \Vdash \exists x \phi \Leftrightarrow \text{there exists } a \in A_w \text{ s.t. } w \Vdash \phi[x:=a]$$

$$w \Vdash \phi \rightarrow \psi \Leftrightarrow \text{for all } v \leq w \quad v \Vdash \phi \cdot v \text{ implies } v \Vdash \psi \cdot v$$

$$w \Vdash \forall x \phi \Leftrightarrow \text{for all } v \leq w \text{ and } a \in A_v \quad v \Vdash (\phi \cdot v)[x:=a]$$

$1 \Vdash \text{Hesperus} = \text{Phosphorus}$

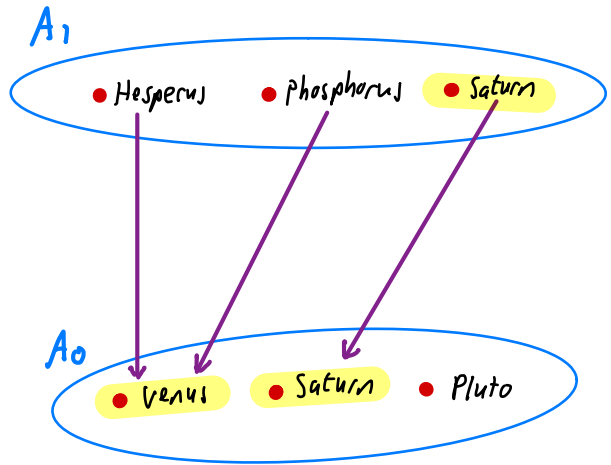
$0 \Vdash \text{Hesperus} \cdot 0 = \text{Phosphorus} \cdot 0$

$1 \Vdash \neg (\text{Hesperus} = \text{Phosphorus})$

$1 \Vdash H = P \vee \neg (H = P)$

$1 \Vdash \neg \neg (H = P)$

$$W = \begin{matrix} 1 \\ | \\ 0 \end{matrix}$$



 The planet predicate

$\neg \phi := \phi \rightarrow \perp$

$w \Vdash \neg \phi \Leftrightarrow \forall v \leq w. v \nVdash \phi$

$w \Vdash \neg \neg \phi \Leftrightarrow \forall v \leq w. \neg \neg \exists u \leq v. u \Vdash \phi$

Meta theorems for Kripke semantics

Monotonicity If $w \Vdash \phi$ and $v \leq w$ then $v \Vdash \phi$.

Soundness If ϕ is provable in intuitionistic predicate logic then for all Kripke models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$.

Completeness If, for all Kripke models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$ then ϕ is provable in intuitionistic predicate logic.

Completeness is not provable in a constructive meta-theory.

To achieve constructive completeness, we need more general notion of model.

Coverage base

a.u.a. base for a
Grothendieck topology
on a poset

A coverage base on a poset $\mathbb{W}$ is a relation $\triangleright$ relating pairs $C \triangleright w$, where $w \in \mathbb{W}$ and $C \subseteq \downarrow w$, satisfying:

$$\downarrow w := \{v \in \mathbb{W} \mid v \leq w\}$$

$$\triangleright \subseteq \sum_{w \in \mathbb{W}} \mathcal{P}(\downarrow w)$$

Reflexivity $\{w\} \triangleright w$

Transitivity If $C \triangleright w$ and, for all $v \in C$, $C_v \triangleright v$ then $\bigcup_{v \in C} C_v \triangleright w$.

Stability If $C \triangleright w$ and $w' \leq w$ then there exists $D \triangleright w'$ such that, for all $v' \in D$ there exists $v \in C$ such that $v' \leq v$.

Example coverage bases

① The identity coverage $\{w\} \triangleright w$ on any poset W

② For any topological space T , define

$$W = \mathcal{O}(T)$$

(open sets)

$$v \leq w \iff v \subseteq w$$

$$C \triangleright w \iff \bigcup C = w$$

(C covers w)

Exercise Prove that ② satisfies properties of a cover base!

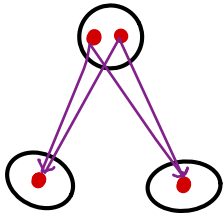
A separated Model is given by

- A partial order $(W, \leq)$ of worlds with coverage base $\triangleright$
- To every $w \in W$ a set A_w
- Transition functions $(t_{vw} : A_w \rightarrow A_v)_{v \leq w}$ s.t. $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- $C \triangleright w \Rightarrow \forall a, b \in A_w (\forall v \in C, t_{vw}(a) = t_{vw}(b)) \Rightarrow a = b$. ($\triangleleft$ -separated)
- For every predicate P of arity k subsets $P_w \subseteq (A_w)^k$ satisfying
 - $\forall \vec{a} \in A_w^k (\vec{a} \in P_w \Rightarrow \forall v \leq w, t_{vw}(\vec{a}) \in P_v)$
 - $C \triangleright w \Rightarrow \forall \vec{a} \in A_w^k (\forall v \in C, t_{vw}(\vec{a}) \in P_v) \Rightarrow \vec{a} \in P_w$ (P is $\triangleleft$ -closed)

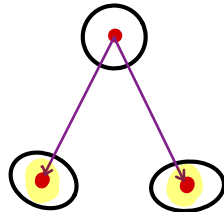
Category-theoretic Formulation

A separated model is given by

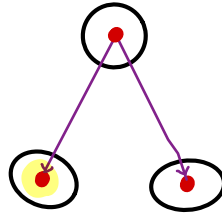
- A partial order $(W, \leq)$ of worlds
- A $\triangleright$ -separated presheaf $A : W^{op} \rightarrow \underline{\text{Set}}$
- For every predicate P of arity k , a $\triangleright$ -closed subpresheaf $P \subseteq A^k$



not separated



not closed



a separated model

$$W = \bigwedge_{0 \ 0'}^1$$

$$\{0, 0'\} \triangleright 1$$

$$\{0\} \not\triangleright 1$$

$$\{0'\} \not\triangleright 1$$

New forcing clauses

$w \Vdash \phi \vee \psi \iff$ there exists a cover $C \triangleright w$ such that,
for every $v \in C$, $v \Vdash \phi \cdot v$ or $v \Vdash \psi \cdot v$

$w \Vdash \perp \iff \emptyset \triangleright w$

$w \Vdash \exists x \phi \iff$ there exists a cover $C \triangleright w$ such that,
for every $v \in C$,
there exists $a \in \underline{A}_v$ s.t. $v \Vdash (\phi \cdot v)[x := a]$

N.B.

- The clauses for predicates, $=$, $\wedge$, $\rightarrow$, $\forall$ are as for Kripke semantics

For our example coverage bases

(1) The identity coverage base $\{w\} \triangleright w$ on any poset W

Recovers Kripke semantics

(2) The topological cover relation for $W = \mathcal{O}(T)$.

Recovers topological semantics of intuitionistic logic

Meta theorems for coverage semantics

Monotonicity If $w \Vdash \phi$ and $v \leq w$ then $v \Vdash \phi$.

Cover property If $C \supseteq w$ satisfies, for all $v \in C$, $v \Vdash \phi$,
then $w \Vdash \phi$.

Soundness If ϕ is provable in intuitionistic predicate logic
then for all separated models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$.

Completeness If, for all separated models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$
then ϕ is provable in intuitionistic predicate logic

Completeness is now provable in a constructive meta-theory!

Intuitionistic sequent Calculus

Predicate symbols $P(x_1, \dots, x_n)$ ($n \geq 0$)

No function symbols (in particular no constants) (for brevity of exposition)

Equality $t_1 = t_2$

[The only terms are variables, but we use $t, t_1, \dots$ as meta-variables in places where arbitrary terms would be allowed in presence of function symbols.]

Sequent: $\Theta; \Gamma \Rightarrow \psi$

where Θ is finite set of variables

Γ is finite set of Θ -formulas, ψ is Θ -formula

A Θ -formula is a formula whose free variables are contained in Θ .

Rules

$$\frac{}{\Theta ; \Gamma, P(\vec{t}) \Rightarrow P(\vec{t})}$$

$$\frac{\Theta ; \Gamma, \phi, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \wedge \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \wedge \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \wedge \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \wedge \psi'$$

$$\frac{\Theta ; \Gamma, \phi \Rightarrow \psi \quad \Theta ; \Gamma, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \vee \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \vee \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \vee \psi'$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \vee \psi'$$

$$\frac{}{\Theta ; \Gamma \ni \perp \Rightarrow \psi}$$

$$\frac{\Theta ; \Gamma, \phi' \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \phi}{\Theta ; \Gamma \ni \phi \rightarrow \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \rightarrow \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma, \psi \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \rightarrow \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \rightarrow \psi'$$

$$\frac{\Theta \wp x; \Gamma, \phi \Rightarrow \psi}{\Theta; \Gamma \ni \exists x \phi \Rightarrow \psi}$$

$$\frac{\Theta; \Gamma' \Rightarrow \psi[t/x]}{\Theta; \Gamma \Rightarrow \exists x \psi}$$

$$\frac{\Theta; \Gamma', \phi[t/x] \Rightarrow \psi}{\Theta; \Gamma \ni \forall x t \Rightarrow \psi}$$

$$\frac{\Theta \wp x; \Gamma \Rightarrow \psi}{\Theta; \Gamma \Rightarrow \forall x \psi}$$

$$\frac{\Theta; \Gamma \Rightarrow \psi[s/z]}{\Theta; \Gamma \ni s=t \Rightarrow \psi[t/z]}$$

$$\frac{\Theta; \Gamma \Rightarrow \psi[t/z]}{\Theta; \Gamma \ni s=t \Rightarrow \psi[s/z]}$$

$$\frac{}{\Theta; \Gamma \Rightarrow t=t}$$

$$\frac{\Theta; \Gamma \Rightarrow \phi \quad \Theta; \Gamma, \phi \Rightarrow \psi}{\Theta; \Gamma \Rightarrow \psi} \text{ cut}$$

Define $\Theta; \Gamma \vdash \psi$ $:\Leftrightarrow$ the sequent $\Theta; \Gamma \Rightarrow \psi$ is provable

$\Theta; \Gamma \vdash_{cf} \psi$ $:\Leftrightarrow$ the sequent $\Theta; \Gamma \Rightarrow \psi$ is cut-free provable.

Lemma (admissible rules with/without cut)

- (identity) $\Theta; \Gamma, \psi \vdash_{cf} \psi$
- (weakening) $\Theta; \Gamma \vdash \psi \Rightarrow \Theta, \theta'; \Gamma, \Gamma' \vdash \psi$
 $\Theta; \Gamma \vdash_{cf} \psi \Rightarrow \Theta, \theta'; \Gamma, \Gamma' \vdash_{cf} \psi$

Proofs Straightforward inductions.

Theorem (Cut admissibility) $\Theta; \Gamma \vdash_{cf} \phi$ and $\Theta; \Gamma, \phi \vdash_{cf} \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

We shall give a semantic proof, by establishing

Soundness $\Theta; \Gamma \vdash \psi \Rightarrow \Theta; \Gamma \models \psi$

Cut-free completeness $\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

where $\models$ is semantic entailment w.r.t. separated models.

The proof will be carried out in a constructive meta-theory, so (in principle) it's possible to extract from the proof an algorithm that maps derivations with cuts to cut-free derivations.

Related work: Beth, Veldman et.al., Palmgren, Avigad

Semantic entailment

$$\Theta; \Gamma \models \psi$$

$:\Leftrightarrow$

for every poset W and coverage base $\triangleright$ on W ,

for every separated model $(A: W^{op} \rightarrow \underline{\text{set}}, \{p_w\}_w)$,

for every $w \in W$,

for every $\rho: \Theta \rightarrow A_w$

if, for every $\phi \in \Gamma$, $w \Vdash \phi[\rho]$,

then $w \Vdash \psi[\rho]$

$\forall w$

$\Theta; \Gamma \models_m \psi$

Proof of soundness: $\Theta; \Gamma \vdash \psi \Rightarrow \Theta; \Gamma \models \psi$

Consider any m , (i.e., $\mathbb{W}, \triangleright$ and separated model $A, \dots$)

We prove by ind. on derivation of $\Theta; \Gamma \vdash \psi$ that $\Theta; \Gamma \models_m \psi$ holds

Exercise!

Proof of cut-free completeness: $\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

In Part 1, we consider just the case of intuitionistic propositional logic

- All predicate symbols have arity 0.
(Predicates of arity 0 are atomic propositions.)
- No variables, no quantifiers $\forall, \exists$
- No equality $=$

We build a specific separated model $\mathcal{M}_{ctx}$

The modal M_{ctx} (propositional logic)

$W_{ctx} := \{ \Gamma \mid \Gamma \text{ a finite set of formulas} \}$ (The set of contexts)

$\Gamma' \leq \Gamma : \Leftrightarrow \Gamma' \supseteq \Gamma$

$C \triangleright_{ctx} \Gamma : \Leftrightarrow \frac{C}{\Gamma}$ is a derivable context rule

$A_{\Gamma} := \emptyset$

$() \in P_{\Gamma} : \Leftrightarrow \Gamma \vdash_{cf} P \quad . \quad (\Gamma \Vdash P : \Leftrightarrow \Gamma \vdash_{cf} P)$

○ Corrected from lecture. (Thanks to Dominik Wehr)

Context rules

$$\frac{\Gamma, \phi, \phi'}{\Gamma \ni \phi \wedge \phi'}$$

$$\frac{\Gamma, \phi \quad \Gamma, \phi'}{\Gamma \ni \phi \vee \phi'}$$

$$\frac{}{\Gamma \ni \perp}$$

$$\frac{\Gamma, \phi'}{\Gamma \ni \phi \rightarrow \phi'} \quad \Gamma \vdash_{cf} \phi$$

For any context Γ and set of contexts C ,

$\frac{C}{\Gamma}$ is a derivable context rule if there exists a derivation tree of contexts with conclusion Γ , in which every leaf is in C .

Lemma Suppose $\frac{C}{\Gamma}$ is a derivable context rule.

- (Weakening) For any Γ'' ,

$$\frac{\{\Gamma', \Gamma'' \mid \Gamma' \in C\}}{\Gamma, \Gamma''}$$

is a derivable context rule

The stability
of $\triangleright_{\text{ctx}}$
follows.

- (Sequencing) For any ψ ,

$$\frac{\{\Gamma' \Rightarrow \psi \mid \Gamma' \in C\}}{\Gamma \Rightarrow \psi}$$

is a cut-free derivable sequent rule.

The $\triangleright$ -closed
property of
 M_{ctx}
follows

Main lemma $\mathcal{M}_{ctx}$ enjoys the following.

For any $\Gamma \in \mathcal{W}_{ctx}$ and formula ϕ

(1) if $\phi \in \Gamma$ then $\Gamma \Vdash \phi$.

(2) if $\Gamma \Vdash \phi$ then $\Gamma \vdash_{cf} \phi$.

Proof of cut-free completeness: $\Gamma \models \psi \Rightarrow \Gamma \vdash_{cf} \psi$

Suppose $\Gamma \models \psi$.

In particular $\Gamma \models_{\mathcal{M}_{ctx}} \psi$. $\otimes$

By ① of main lemma, for every $\phi \in \Gamma$, $\Gamma \Vdash \phi$.

By $\otimes$, $\Gamma \Vdash \psi$.

By ② of main lemma, $\Gamma \vdash_{cf} \psi$.

□

Main lemma M_{ctx} enjoys the following.

For any $\Gamma \in W_{ctx}$ and formula ϕ

(1) if $\phi \in \Gamma$ then $\Gamma \Vdash \phi$.

(2) if $\Gamma \Vdash \phi$ then $\Gamma \vdash_{cf} \phi$.

Proof

(1) and (2) are proved simultaneously by induction on ϕ .

□

Case $\phi \equiv \phi_1 \rightarrow \phi_2$

(1) $\phi_1 \rightarrow \phi_2 \in \Gamma \Rightarrow \Gamma \Vdash \phi_1 \rightarrow \phi_2$

Suppose $\Gamma' \supseteq \Gamma$ and $\Gamma' \Vdash \phi_1$,

By i.h. (2) $\Gamma' \vdash_{cf} \phi_1$

By $\bigcirc$, $\{\Gamma', \phi_2\} \supseteq_{ctx} \Gamma'$

By i.h. (1), $\Gamma', \phi_2 \Vdash \phi_2$

By closing property $\Gamma' \Vdash \phi_2$.

Case $\phi \equiv \phi_1 \rightarrow \phi_2$

$$(2) \quad \Gamma \Vdash \phi_1 \rightarrow \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \rightarrow \phi_2$$

By i.h. (1) $\Gamma, \phi_1 \Vdash \phi_1$

By $\bigcirc$, $\Gamma, \phi_1 \Vdash \phi_2$

By i.h. (2) $\Gamma, \phi_1 \vdash_{cf} \phi_2$

Hence $\Gamma \vdash_{cf} \phi_1 \rightarrow \phi_2$ by $(\rightarrow R)$.

Case $\phi \equiv \phi_1 \vee \phi_2$

(1) $\phi_1 \vee \phi_2 \in \Gamma \Rightarrow \Gamma \Vdash \phi_1 \vee \phi_2$

We have $\{(\Gamma, \phi_1), (\Gamma, \phi_2)\} \triangleright_{ctx} \Gamma$

By i.h. (1), $\Gamma, \phi_1 \Vdash \phi_1$ and $\Gamma, \phi_2 \Vdash \phi_2$.

Hence $\Gamma \Vdash \phi_1 \vee \phi_2$.

Case $\phi \equiv \phi_1 \vee \phi_2$

$$(2) \quad \Gamma \Vdash \phi_1 \vee \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \vee \phi_2$$

By $\circ$, there is $C \triangleright_{ctx} \Gamma$ such that

for every $\Gamma' \in C$, $\Gamma' \Vdash \phi_1$ or $\Gamma' \Vdash \phi_2$.

So for every $\Gamma' \in C$, $\Gamma' \vdash_{cf} \phi_1$ or $\Gamma' \vdash_{cf} \phi_2$, by i.h (2).

So for every $\Gamma' \in C$, $\Gamma' \vdash_{cf} \phi_1 \vee \phi_2$, by ($\vee R$) rules.

Since $\frac{C}{\Gamma}$ is admissible, $\Gamma \vdash_{cf} \phi_1 \vee \phi_2$, by sequencing property.



Sieve

In a coverage (rather than coverage base) we only allow covers $C \triangleright w$, where C is a sieve on w .

A set $C \subseteq W$ is a sieve on $w \in W$ if

- $C \subseteq \downarrow w$
- $C = \downarrow C$ (C is down closed)

If C is a sieve on w and $v \leq w$ then

$$C \cdot v := \{u \in C \mid u \leq v\}$$

is a sieve on v .

$$w \mapsto \text{Sieves}(w)$$

defines a presheaf

$$W^{op} \rightarrow \underline{\text{Set}}$$

Coverage

A coverage is a relation $C \triangleright w$ between $w \in W$ and sieves C on w satisfying

Reflexivity $\downarrow w \triangleright w$

Transitivity If $C \triangleright w$ and S is a sieve on w satisfying

$$\forall v \in C, S \cdot v \triangleright v$$

then $S \triangleright w$.

Stability If $C \triangleright w$ and $v \leq w$ then $C \cdot v \triangleright v$.

Proposition (Properties of coverage)

Let $\triangleright$ be a coverage. Then $\triangleright$ satisfies

Monotonicity If $C \triangleright w$ and S is a sieve on w then

$$C \subseteq S \Rightarrow S \triangleright w$$

Intersection If $C \triangleright w$ and $C' \triangleright w$ then $C \cap C' \triangleright w$

Proof hints For monotonicity use reflexivity + transitivity.

For intersection use transitivity + stability.

Coverages vs. Coverage bases

Proposition

(1) If $\triangleright$ is a coverage base then

$$C \triangleright w \quad :\Leftrightarrow \quad \exists B \subseteq C, B \triangleright w \text{ and } C = \downarrow B$$

is a coverage. We say $\triangleright$ is the coverage generated by $\triangleright$.

(2) Every coverage $\triangleright$ is generated by the coverage base

$$B \triangleright w \quad :\Leftrightarrow \quad \downarrow B \triangleright w.$$

Proof Exercise!

The $\neg\neg$ -dense coverage

On any poset W define

$$C \triangleright_{\neg\neg} w \quad :\Leftrightarrow \quad C \text{ is a sieve on } w \text{ and} \\ \forall v \leq w \quad \neg\neg \exists u \leq v \quad u \in C.$$

Proposition $\triangleright_{\neg\neg}$ is a coverage.

Proof Exercise.

$\neg\neg$ Dense $\Rightarrow$ Classical

Proposition Using the coverage $\triangleright_{\neg\neg}$, $\forall w \in W$ with $\neg\neg\phi \rightarrow \phi$

Proof Suppose $w \Vdash \neg\neg\phi$

we show $\{u \leq w \mid u \Vdash \phi\} \triangleright_{\neg\neg} w$. Then $w \Vdash \phi$ by cover property.

$\{u \leq w \mid u \Vdash \phi\}$ is a sieve by monotonicity of $\Vdash$

Suppose $v \leq w$. We need to show $\neg\neg \exists u \leq v, u \Vdash \phi$.

Suppose for $\perp$ that $\neg \exists u \leq v, u \Vdash \phi$.

Then $v \Vdash \neg\phi$.

By $\circ$, $v \Vdash \perp$. So $\emptyset \triangleright_{\neg\neg} v$. So $\forall v' \leq v$ $\neg\neg \exists u' \leq v, u' \Vdash \phi$. I.e. $\perp$.

□